INDEPENDENT RESEARCH REPORT · SALMON ECOLOGY

Life-Stage Water Temperature and Salmon Returns to Issaquah Creek, 1997–2025

An observational test of five prespecified hypotheses using a locally calibrated temperature reconstruction

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Abstract

Coho returns to Issaquah Creek, Washington, vary by nearly a hundredfold between years and Chinook returns by roughly sevenfold, and elevated water temperature is widely treated as a stressor for migrating and rearing salmon. This study asked whether water temperature during biologically specific life stages is associated with that year-to-year variation, and whether the hydrologic pathway usually assumed to produce warm water holds in this watershed. Because no continuous daily water-temperature record exists for Issaquah Creek at the resolution these questions require, daily temperature for 1995–2025 was reconstructed by ridge regression from regional air temperature and seasonal terms, calibrated against 431 King County grab samples and validated by leave-one-year-out cross-validation; held-out error inside the three biological windows was 0.70–0.97 °C, 36–47% better than a monthly-average baseline. Five hypotheses in two prespecified families — three relating life-stage temperature to adult returns, two relating snowpack to flow and flow to temperature — were tested with Spearman rank correlation, 100,000-permutation inference, and Holm correction. The three salmon hypotheses used n = 29 return years (1997–2025); the two physical-mechanism analyses used up to 31 years of environmental record (1995–2025). The analysis protocol was frozen before any of these life-stage tests was run, though earlier project phases had already examined more general annual temperature-and-return relationships.

No salmon-temperature hypothesis met the prespecified support threshold. Chinook returns showed a moderate negative association with migration-window temperature ($\rho = -0.400$; raw $p = 0.032$; Holm $p = 0.095$) that kept its direction and approximate magnitude across all seven prespecified robustness checks, including all 29 leave-one-year-out reruns, but remains statistically inconclusive. Both Coho estimates were imprecise, with bootstrap intervals spanning zero. In the mechanism family, April snow-water equivalent at Stampede Pass was not informative for Issaquah Creek late-summer flow ($\rho = +0.137$), while higher summer flow was associated with cooler modeled water ($\rho = -0.417$; Holm $p = 0.039$), so the proposed chain was not observed intact. The principal limitations are a 29-year sample with limited power, a modeled rather than measured exposure, and unmeasured confounders including hatchery release numbers. The contribution is methodological as much as ecological: a flow-independent local temperature reconstruction combined with life-stage exposure windows and a frozen analysis protocol, reported without revision after the results were known. No causal claim is made.

1. Introduction and study objectives

Each fall, Chinook and Coho salmon return from the Pacific Ocean through Puget Sound, the Ballard Locks, and Lake Washington to Issaquah Creek, where a portion are trapped and counted at the Issaquah Salmon Hatchery. That counting has continued long enough to produce a usable long-term record, and the first thing the record shows is instability: between 1997 and 2025, Coho returns ranged from about 360 fish to more than 35,000 — a factor of roughly 98 — and Chinook from roughly 1,800 to 13,400, a factor of about 7.5. Neighbouring years can differ severalfold: Coho returns changed by more than tenfold between adjacent years three times in this record, and the largest adjacent-year change in Chinook was about 4.5-fold.

Many factors are thought to contribute to that variability — ocean survival, harvest, hatchery release numbers, predation at the Locks, habitat condition, and cohort age structure — and almost none could be measured here. What could be measured was the freshwater environment the fish pass through. Elevated water temperature is a well-documented stressor for Pacific salmon: it raises metabolic demand, increases susceptibility to disease, and can render otherwise suitable migration and rearing habitat unusable (McCullough, 1999; U.S. EPA, 2003; Richter & Kolmes, 2005). A second, hydrologic idea motivated the study in parallel: that reduced winter snowpack lowers late-summer streamflow, and that a creek carrying less water warms more readily. Both ideas were recorded in the project's planning documents before the life-stage analysis reported here was specified.

1.1 The local evidence gap

Testing those ideas at Issaquah Creek runs into a measurement problem. The USGS gauge on the creek reports discharge only. King County measures water temperature at station 0631, but as irregular grab samples rather than a continuous series. **No continuous daily local temperature record exists at the resolution required to construct biologically defined exposure windows** — a narrower statement than saying no temperature data exist, since multi-decade Issaquah temperature records of a different form are reported in recent work (Quinn, Ray, & Tillotson, 2026). The form, resolution, and availability of that record could not be determined from the published abstract alone and it was not obtained for this study, so reconstructing a defensible daily series was a precondition for the biological questions rather than a side task; §7 returns to this as a route worth pursuing.

Prior work establishes the components but not the combination. Run timing for Chinook and Coho in Puget Sound is documented and largely non-overlapping, with Chinook arriving from late July through September and Coho from early September into November (WDFW, Chinook species profile; WDFW,

Lake Washington salmon counts). Wild Coho typically rear in fresh water for about a year before smolting (Anderson et al., 2026; NOAA Fisheries), while Issaquah hatchery Coho are held in rearing ponds for that year and released as yearlings (Friends of the Issaquah Salmon Hatchery, operations page) — a distinction that matters for interpreting the juvenile-rearing hypothesis. Within the Lake Washington basin, warming river temperature has been linked to increased prespawning mortality in sockeye and to lower recovery of radio-tagged sockeye passing the Ballard Locks in warmer years (Newell et al., 2007; Barnett et al., 2020).

The nearest work on this exact system is Quinn, Ray, & Tillotson (2026), which uses the same hatchery, the same two species, and Issaquah Creek temperature records back to 1970. It reports that Chinook and Coho there now spawn earlier in the fall than in past decades despite roughly 1.5–2 °C of warming, attributing the shift to hatchery-driven selection. That study is genuinely adjacent but addresses a different response variable: spawn timing, not the size of the annual adult run, and it does not test life-stage exposure windows against return abundance. A literature search did not identify a study taking that second step for this creek; that is a search result rather than a proof of novelty, and only the published abstract of Quinn et al. was reviewed.

1.2 Objectives and prespecified hypotheses

Correlating an annual average temperature with an annual return count would be the wrong test, because salmon are not exposed to "the year." A Chinook adult ascending the creek in September experiences a specific six-week window; a juvenile Coho experiences a different window two years before it returns as an adult. Averaging across the year mixes the weeks that matter biologically with those that do not. The question was therefore posed as five hypotheses tied to specific windows (Table 1): three relating life-stage temperature to adult returns, and two testing the physical links the snowpack reasoning depends on. Two additional windows (A2 and A4) were prespecified as longer sensitivity versions of A1 and A3 and were never permitted to substitute for the primary windows.

Table 1. The five prespecified hypotheses. Each names a species and life stage, an exposure window, its alignment to a return year, and the direction expected before analysis. A1–A7 are the project's analysis codes.

ID	Hypothesis (as written before testing)	Species / life stage	Exposure window	Alignment	Expected
A1	Warmer water during adult Chinook migration is associated with fewer returning adults.	Chinook, adult migration	Aug 15 – Sep 30	Same year	Negative
A3	Warmer water during adult Coho migration is associated with fewer returning adults.	Coho, adult migration	Sep 15 – Oct 31	Same year	Negative
A5	Warmer water during the Coho juvenile rearing summer is associated with fewer adults two years later.	Coho, juvenile rearing	Jun 1 – Sep 30	Return year − 2	Negative
A6	Greater April snowpack is associated with greater late-summer streamflow.	— (physical link)	Apr 1 SWE → Jul–Sep flow	Same year	Positive
A7	Greater summer streamflow is associated with cooler summer water temperature.	— (physical link)	Jun–Sep flow → Jun–Sep temp.	Same year	Negative

Source: Hypothesis definitions transcribed from the project's frozen Phase 7 analysis protocol (version 1.1, 23 August 2026), §3. **Table prepared by** S. Tian, 2026, authored in HTML/CSS and rendered with python-docx 1.2.0 under Python 3.13.7 (Word version) and headless Chrome (PDF version).

2. Data and temperature reconstruction

All data are public secondary sources (Table 2); no field data were collected. The advantage is that the records are long, independently maintained, and re-downloadable by anyone checking the work; the cost is that each measurement was collected for purposes other than this analysis. Study years are 1997–2025 for salmon outcomes and 1995–2025 for the temperature reconstruction, the earlier start being required by the two-year lag in A5.

Table 2. Data sources and their role in the analysis. Each dataset was downloaded once and stored unmodified with a checksum. Full citations and retrieval URLs appear in the Bibliography.

Source	Variable used	Role	Coverage
WDFW Hatchery Adult Salmon Returns	Adult trap counts, by species and origin	Outcome variable	1997–2025
King County station 0631	Grab-sample water temperature	Calibration target for the temperature model	1995–2025
NOAA NCEI GHCN-Daily, station USW00024233	Daily air temperature (min/max)	Predictor in the temperature model	1994–2025
USGS gauge 12121600	Daily mean discharge	Flow variable; predictor in T1 only	1994–2025
NRCS SNOTEL station 788 (Stampede Pass)	April 1 snow-water equivalent	Regional snowpack indicator (A6)	1997–2025

Source: Compiled from the project's source register, which records the query URL, access date, coverage, and licence terms for every dataset. **Table prepared by** S. Tian, 2026, authored in HTML/CSS and rendered with python-docx 1.2.0 under Python 3.13.7 (Word version) and headless Chrome (PDF version).

2.1 Outcome definition and sample

The outcome is **total adults**: hatchery-origin and wild-origin adult counts summed within a species and return year, with jacks (early-maturing males) excluded from the primary analysis and retained for a

robustness check. Origins are pooled because an origin-comparability audit across the record was not completed, and the protocol forbids introducing an origin split after results are known. No missing count was imputed; a year lacking a usable value was dropped. After joining exposure to returns, each salmon hypothesis had exactly 29 usable return years. Because the counts are administrative records maintained over three decades, changes in counting procedure across years cannot be ruled out; no comparability check was performed, and this is treated as a limitation (§6).

2.2 Reconstructing a daily temperature series

Over thirty-one years, King County collected 431 temperature measurements on 415 distinct dates at station 0631 — roughly one reading every twenty-seven days, unevenly spaced and concentrated in daytime hours. Averaging a handful of such samples into a "summer temperature" would measure sampling luck as much as climate: a year whose few readings happened to fall during a heat wave would appear warm. A model was therefore built to predict creek temperature on every day from 1995 to 2025 using inputs with complete daily coverage.

Two ridge-regression models were fitted. **T1** uses daily air temperature (midpoint and range), 3-, 7-, and 30-day trailing air-temperature means, annual and semiannual seasonal terms, and streamflow. **T2** uses the same inputs without streamflow. T2 was designated primary for a logical rather than a performance reason: two analyses (A7 and the secondary model in §4.3) test streamflow against temperature, so a temperature variable built partly from streamflow would guarantee part of its own answer. Excluding flow removes that circularity.

Validation used leave-one-year-out cross-validation — holding out an entire year of grab samples, fitting on the remaining thirty, predicting the held-out year, and repeating thirty-one times. Whole years rather than random samples were held out because samples within a year share weather and would otherwise leak into their own prediction. One caveat is stated explicitly: the ridge penalty was selected from {0.01, 0.1, 1, 10, 100} using the same leave-one-year-out structure later reported as validation, and both models selected 10. Reusing the hold-out for tuning and reporting makes the quoted error mildly optimistic by an amount that was not quantified.

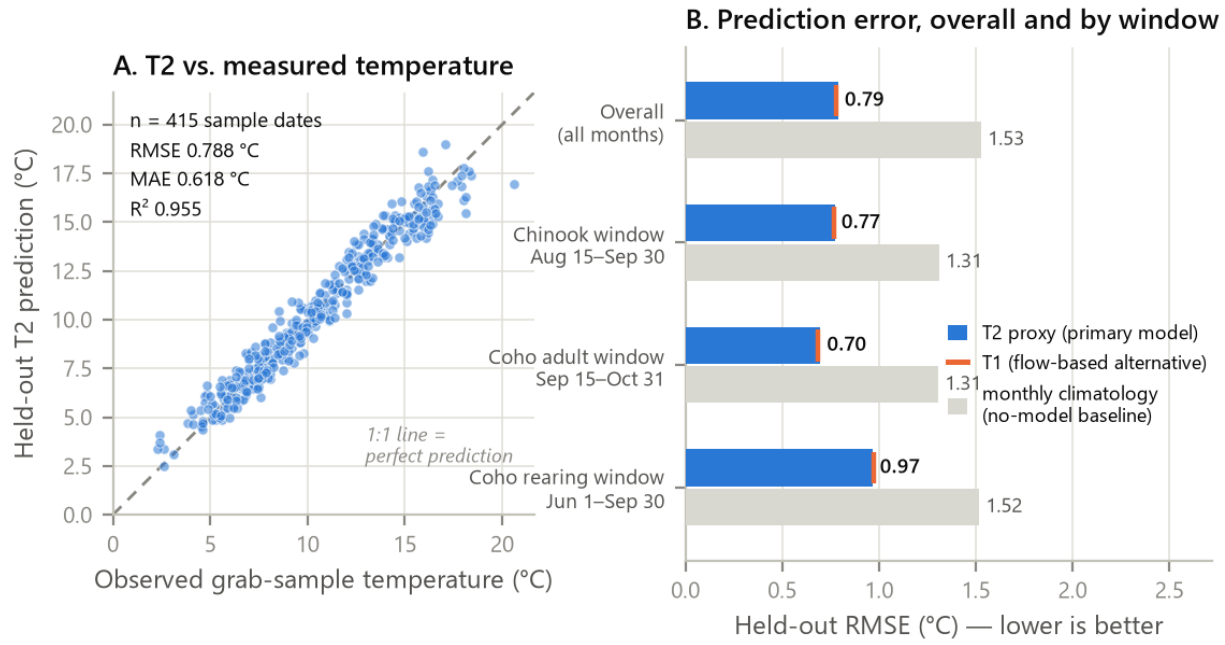


Figure 1. Validation of the reconstructed temperature series. (A) Each point is one grab-sample date plotted against the model's prediction for that date while its entire year was held out of fitting. **(B)** Held-out error overall and within the three biological windows, compared with a monthly-climatology baseline. The orange tick marks T1, the flow-inclusive alternative; it falls almost on T2, which is why excluding flow costs little accuracy.

Figure created by S. Tian, 2026, using Python 3.13.7 with matplotlib 3.11.1. **Data:** King County Water Quality station 0631; NOAA NCEI GHCN-Daily station USW00024233; USGS gauge 12121600 (T1 only). See Bibliography for full source citations.

Across all months, held-out RMSE was 0.788 °C for T2 and 0.781 °C for T1, against 1.527 °C for the monthly-climatology baseline. The overall R^2 of 0.955 is not the figure by which this model should be judged: most of that variance is the seasonal cycle, which any model given air temperature and seasonal terms ought to reproduce. The informative comparison is inside the windows the hypotheses use, where the seasonal signal is largely removed. There T2 achieved held-out RMSE of 0.77 °C (Chinook window), 0.70 °C (Coho adult), and 0.97 °C (Coho rearing), improving on the window-specific climatology baseline by 41%, 47%, and 36% respectively (Figure 1B). Excluding streamflow cost 0.9% in overall RMSE.

BOUNDARY ON HOW THESE VALUES MAY BE USED

The model is calibrated to King County grab samples, which are spot measurements taken largely during daytime hours. Its output is therefore an estimate of a *spot temperature*, not a daily mean and not a daily maximum. A seven-day average of model output is **not** the regulatory 7-day average daily maximum (7DADMax) used in water-quality standards, and threshold-exceedance counts computed from it were designated exploratory and support no conclusion in this report.

2.3 Extrapolation audit and exposure construction

A model calibrated on one range of conditions becomes unreliable outside it. All 11,323 daily predictions were audited and flagged where any input fell outside the range spanned by the calibration samples. Flagging indicates that the predictors lie outside the calibration domain, not that the prediction is known to be wrong. For T2, 248 days (2.19%) were flagged, of which 54 fell in June–October (1.14% of days in those months). The most extreme case is 28 June 2021, during the Pacific Northwest heat dome, where the largest air-temperature input sat 26.3% beyond the calibration range. Flagged years were retained in the primary analysis, with an exclusion sensitivity prespecified (§4.4), because deleting them would have silently changed the sample.

For each hypothesis and year, daily T2 values were averaged across that hypothesis's window, producing one exposure value per year. "Temperature" throughout the Results means this *window-mean modeled value*. The three windows sit at distinct levels — the fall Coho window averages about 11.7 °C, roughly three degrees cooler than the two summer windows — and all three drift upward across the record, most visibly after about 2012. That drift makes temperature and calendar year correlated, which is addressed directly in §4.4.

3. Statistical analysis plan

With thirty-one years of temperature and twenty-nine years of counts, many window-lag-species combinations could have been tried, and with that much freedom a small dataset will eventually yield something that looks significant. The full analysis plan was therefore written into a protocol document and frozen before any of the five hypotheses reported here was tested. That freeze covers this life-stage analysis specifically: earlier phases of the project had already examined more general annual temperature-and-return relationships, a limitation returned to in §3.2.

The protocol fixed every predictor, outcome, window, and lag (Table 1); the primary statistic; the inference procedure; the multiplicity families; every robustness check; and the rule for what may be called supported. The primary statistic is the **Spearman rank correlation**, chosen because it assumes neither linearity nor normality and is robust to the extreme high-return years in the record. Significance is assessed by a **two-sided permutation test with 100,000 reshuffles** under fixed random seeds, so the values are exactly reproducible; expected direction was prespecified for reporting, but the test itself remained two-sided. Interval estimates come from **10,000 paired bootstrap resamples**. A supporting ordinary-least-squares regression with HC3 robust standard errors accompanies each primary test and is explicitly subordinate to it.

3.1 Multiplicity families and the support rule

Testing several hypotheses at once gives several chances at a low *p*-value by luck, so the **Holm step-down correction** was applied to hold the family-wise error rate at 5%. Two families were defined in advance: the three salmon-temperature hypotheses (A1, A3, A5) in one, and the two mechanism hypotheses (A6, A7) in the other. The scientific rationale is that they predict different kinds of quantity — one family predicts fish counts, the other predicts a physical property of the creek — so they answer separable questions rather than competing for the same claim.

It should be stated plainly which way that choice cuts. Splitting five tests into families of three and two is **less** stringent than correcting all five together, because Holm's threshold depends on how many tests share a family. Section 4.2 reports what a single five-test family would have produced. The justification is that the families were defined before any result existed, not that splitting is the more conservative option.

THE DECISION RULE, FIXED BEFORE ANY RESULT EXISTED

A relationship may be described as **supported by this observational analysis** only if all three conditions hold: the observed direction matches the direction predicted in advance; the **Holm-adjusted** *p*-value is below 0.05; and the direction does not reverse in the required robustness checks. Otherwise the result is reported as weak, uncertain, or not supported by these data, with the effect estimate and interval shown regardless. **Failing this rule is not evidence that no biological effect exists**; it means these data did not establish one.

3.2 Prespecified robustness and temporal-dependence checks

Each primary test was accompanied by prespecified sensitivity analyses: substituting the flow-inclusive T1 proxy, substituting the longer A2/A4 windows, including jacks in the outcome, excluding years containing extrapolation-flagged days, excluding the highest-influence observation by Cook's distance, and a leave-one-year-out rerun across all 29 years. Because both temperature and returns trend with calendar time, an additional temporal-dependence sensitivity was added by dated amendment before the salmon tests ran: a partial Spearman correlation after removing a linear calendar-year trend from both rank series, plus a circular-shift diagnostic that slides the return series by each possible offset while preserving its internal order.

One qualification belongs with the protocol rather than in the results. Earlier phases of this project had already examined generic annual temperature-and-salmon relationships before the Phase 7 plan was written. The freeze therefore prevented further changes during the life-stage analysis but is **not** a fully prospective preregistration, and is described here as prespecified for Phase 7 rather than preregistered.

4. Results

4.1 Primary salmon-temperature family

All three primary tests (Table 3) used the same 29 return years, the same T2 window-mean predictor, the same Spearman statistic, and the same Holm correction. **None of the three met the prespecified family-level support threshold.** That family-level outcome precedes any interpretation of the individual estimates.

Table 3. Results for all five prespecified hypotheses. ρ is the Spearman rank correlation. Bootstrap intervals are the 95% range from 10,000 paired resamples; raw *p* is the permutation value before correction; Holm *p* is corrected within the hypothesis's own family. The decision rule uses Holm *p*.

ID	Relationship tested	n	ρ	95% bootstrap CI	Raw *p*	Holm *p*	Status
Salmon-temperature family (Holm across A1, A3, A5)							
A1	Chinook adult migration temperature → adult returns	29	**−0.400**	−0.677 to −0.045	0.032	**0.095**	Not supported
A3	Coho adult migration temperature → adult returns	29	−0.044	−0.436 to 0.360	0.821	0.821	Not supported
A5	Coho rearing temperature (return year − 2) → adult returns	29	−0.202	−0.601 to 0.235	0.290	0.580	Not supported
Mechanism family (Holm across A6, A7)							
A6	April 1 SWE (Stampede Pass) → Jul–Sep mean flow	29	+0.137	−0.289 to 0.501	0.477	0.477	Not supported
A7	Jun–Sep flow → Jun–Sep modeled temperature	31	**−0.417**	−0.677 to −0.078	0.020	**0.039**	Supported

Table prepared by S. Tian, 2026, from the frozen Phase 7 result tables; authored in HTML/CSS and rendered with python-docx 1.2.0 under Python 3.13.7 (Word version) and headless Chrome (PDF version). **Data:** WDFW Hatchery Adult Salmon Returns; King County station 0631; NOAA NCEI GHCN-Daily USW00024233; USGS gauge 12121600; NRCS SNOTEL station 788. See Bibliography.

A1 (Chinook) produced the only substantial estimate: ρ = −0.400, moderate and in the predicted direction. Its raw permutation *p* of 0.032 would be conventionally significant, but after Holm correction within its family it is 0.095, which the prespecified rule classifies as not supported. The supporting regression estimated roughly 21.5% fewer fitted adults per one-standard-deviation warmer window (β = −0.242 on log1p counts; HC3 95% CI −0.463 to −0.021). That estimated association is reported to make the magnitude concrete; it does not override the corrected family-level result, and the bootstrap interval excluding zero does not reinstate the finding.

A3 (Coho adults) produced an estimate near zero, ρ = −0.044, with a bootstrap interval from −0.436 to +0.360. The estimate is essentially zero, but the interval is wide enough to encompass moderately negative and moderately positive associations alike. An estimate near zero and evidence of no relationship are different claims; only the first is supported here. **A5 (Coho rearing)** points in the predicted direction but is likewise inconclusive, ρ = −0.202 with an interval spanning zero (Figure 2).

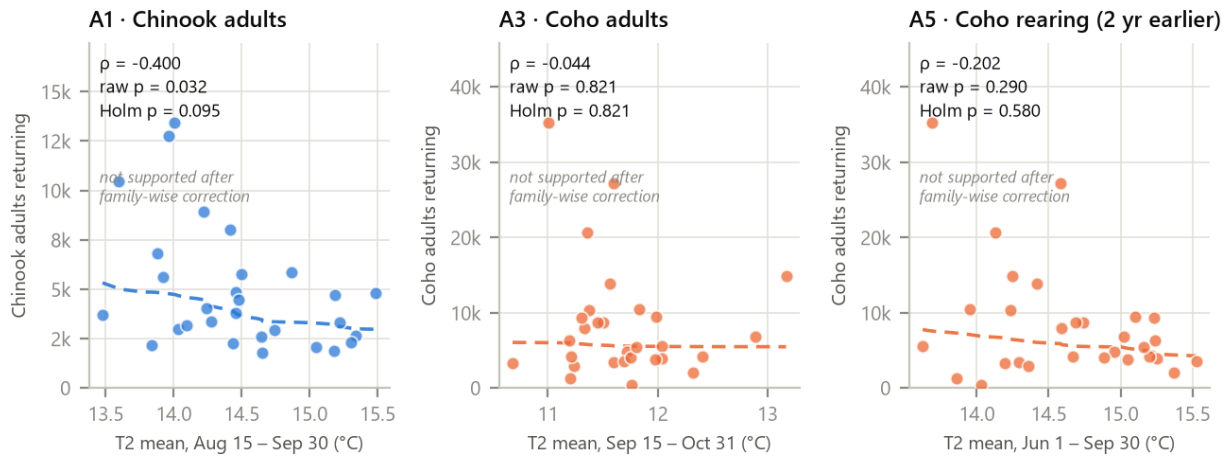


Figure 2. The three primary salmon-temperature hypotheses. One point per return year. The dashed line is a rank-based trend guide indicating the direction ρ measures; it is not a fitted prediction line. All three slope in the predicted direction, but only the Chinook panel shows appreciable strength, and none meets the corrected threshold.

Figure created by S. Tian, 2026, using Python 3.13.7 with matplotlib 3.11.1. **Data:** WDFW Hatchery Adult Salmon Returns; modeled T2 temperature (see Figure 1). See Bibliography.

4.2 Mechanism family

A7 was supported. Years with higher June–September streamflow had cooler modeled water: a one-standard-deviation increase in log streamflow corresponded to about 0.24 °C cooler modeled summer temperature (HC3 95% CI −0.383 to −0.104 °C). It is the only hypothesis in the study meeting its prespecified criterion. The result is an association between measured discharge and *modeled* temperature; it does not demonstrate that increased flow physically cooled the creek. The conventional explanation is that a stream carrying more water has greater thermal mass and shorter residence time, so equivalent solar input raises its temperature less (Caissie, 2006; Stratton Garvin et al., 2022), but heat exchange was not measured, and an annual association cannot separate that mechanism from the simpler possibility that cool, wet summers produce both high flow and low temperature.

A6 was not supported. April 1 snow-water equivalent at Stampede Pass showed no supported relationship with Issaquah Creek late-summer flow (ρ = +0.137, Holm p = 0.477); Figure 3A is close to a shapeless scatter. The conclusion is specific to this proxy: it is a failure of Stampede Pass April SWE to predict Issaquah late-summer flow in this record, not a finding about snowpack in general. One plausible explanation is a scale and regime mismatch. Stampede Pass sits at roughly 1,200 m on the Cascade crest, where seasonal snowpack accumulates and its melt contributes to summer streamflow, whereas the Issaquah Creek basin is a low-elevation watershed of about 58 square miles. Lowland Puget Sound basins are generally classified as rain-dominated rather than snow-dominated (Mauger et al., 2015; Snover et al., 2013), so a Cascade-crest snow gauge is a poor instrument for this creek — though no Issaquah-specific baseflow study was consulted, so the local water balance is not established here.

Taken together, the proposed chain was not observed intact in these data. One of its two physical links was supported and the other was not, so the snowpack → flow → temperature → salmon pathway was not observed intact. A7 concerns water temperature as its outcome and therefore says nothing directly about salmon.

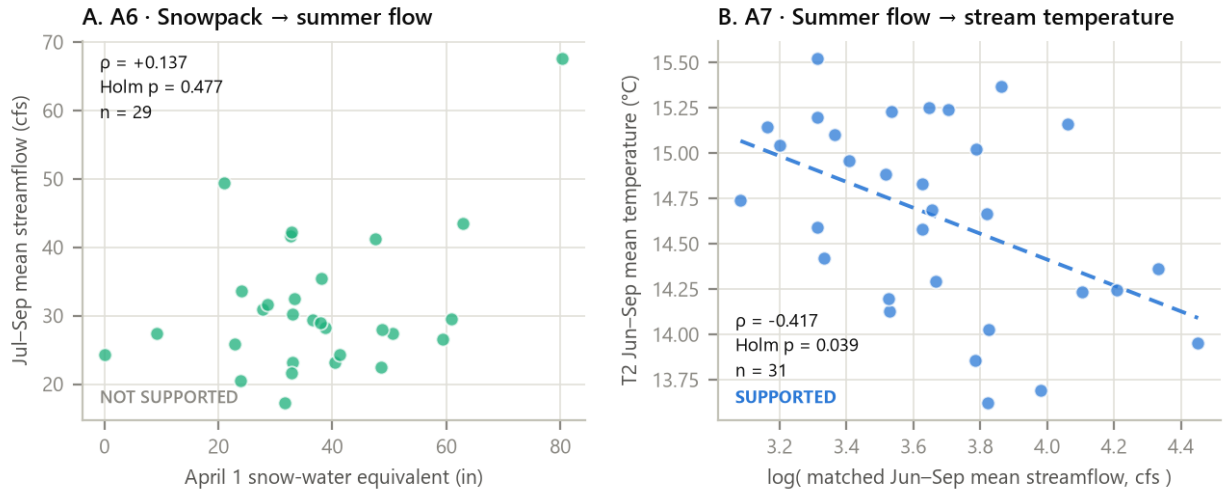


Figure 3. The two physical links tested as the mechanism family. (A) April 1 snow-water equivalent against late-summer streamflow; no trend line is drawn, because fitting one to a null result invites over-reading. **(B)** Summer streamflow (log scale) against modeled summer temperature; higher-flow years had cooler modeled temperature.

Figure created by S. Tian, 2026, using Python 3.13.7 with matplotlib 3.11.1. **Data:** NRCS SNOTEL station 788; USGS gauge 12121600; modeled T2 temperature (see Figure 1). See Bibliography.

Because A7 is the only result crossing the threshold, the multiplicity choice behind it is reported rather than assumed. Under the frozen two-family structure A7's adjusted *p* is 0.039; pooled into a single five-test family it would be 0.099, and no result in the study would meet the threshold. A1 would move from 0.095 to 0.127, A5 from 0.580 to 0.870, A6 from 0.477 to 0.953, and A3 from 0.821 to 0.953 — so A7 is the only classification that depends on the family definition. The two-family split is defensible on the grounds given in §3.1, but it was a judgment call, and the study's single positive result rests on it.

4.3 Secondary combined model

A secondary model entering both window-mean temperature and log matched flow as predictors of returns was prespecified as descriptive and cannot overturn A1 or A3. For Chinook, the standardized temperature coefficient remained negative (β = −0.240; HC3 95% CI −0.478 to −0.001) while flow sat near zero (β = −0.035), with adjusted R^2 = 0.126 and variance inflation factors of 1.006, indicating the two predictors are not competing for the same variance. Removing the most influential year for this combined model (2004) weakened the temperature coefficient to −0.185 with an interval crossing zero. For Coho, neither predictor was distinguishable from zero and adjusted R^2 was negative, meaning the model performed worse than predicting the series mean. Given its secondary protocol status and this

instability, the HC3 $p \approx 0.049$ on the Chinook temperature coefficient is not treated as confirmatory evidence.

4.4 Robustness of the Chinook association

The same prespecified robustness program was run for A1, A3, and A5; because A1 was the only substantial estimate, its results are reported here in full. Its formal status was already settled — not supported after correction — so the question is not whether it is significant but whether it is stable across defensible analytical choices. Every check in Figure 4 was listed in the protocol before any result existed.

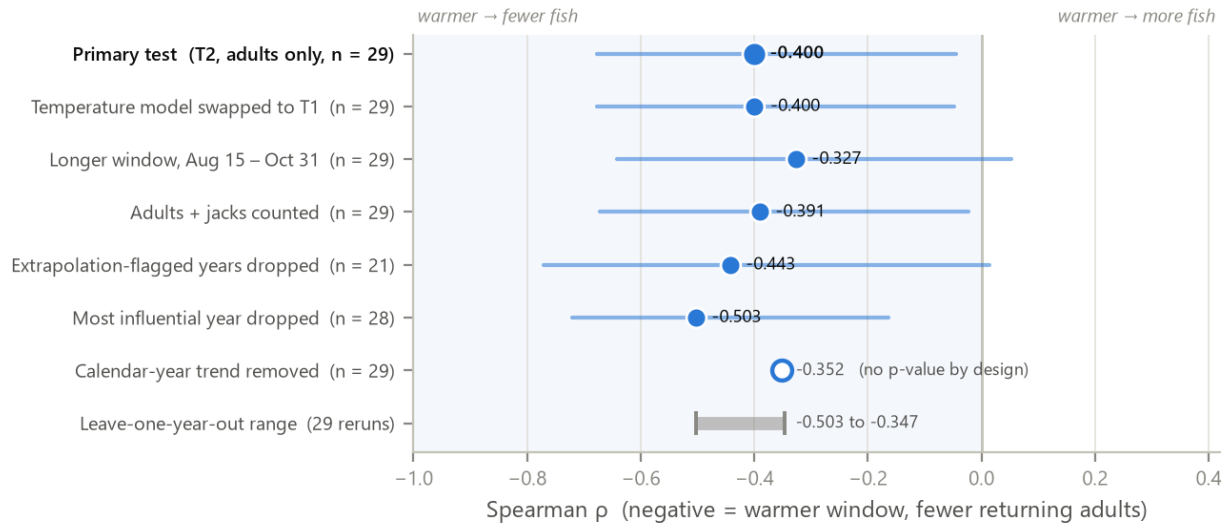


Figure 4. The Chinook association under each prespecified robustness check that yields a correlation estimate. The top row is the primary test; each row below reruns it with one deliberate change. Bars are 95% bootstrap intervals. The detrended row (open marker) carries no interval or *p*-value by design. The bottom row is a range rather than an interval: the full spread of ρ across 29 reruns each omitting one year. The circular-shift diagnostic of §4.4 is not shown, because it returns a shift *p*-value rather than a correlation. No check reversed the sign.

Figure created by S. Tian, 2026, using Python 3.13.7 with matplotlib 3.11.1. **Data:** WDFW Hatchery Adult Salmon Returns; modeled T1 and T2 temperature (see Figure 1). See Bibliography.

The pattern is consistent. Substituting the flow-inclusive T1 proxy left ρ at −0.400; the longer Aug 15 – Oct 31 window weakened it to −0.327; including jacks gave −0.391; excluding the eight years containing extrapolation-flagged days — more than a quarter of an already small sample — gave −0.443; and removing the highest-influence year for this test (1999, a different year from the 2004 point that drives the secondary model in §4.3) gave −0.503. Across all 29 leave-one-year-out reruns ρ ranged from −0.503 to −0.347 with no sign changes, indicating that no single year generates the association. Robustness of this kind increases confidence that the estimate is not an artifact of one analytical choice; it does not establish causality and does not remedy low statistical power.

The temporal-dependence check is the most demanding. Removing a linear calendar-year trend from both rank series weakened A1 from −0.400 to −0.352, about 12% smaller with the direction retained. The defensible inference is narrow: **a shared linear calendar-year trend does not fully account for the rank association.** Nonlinear time trends, cohort effects, ocean regime shifts, and other time-varying confounders remain entirely possible. The residual return series retained lag-1 autocorrelation of 0.54, which triggered the prespecified circular-shift diagnostic; its exact two-sided value was 1/29 = 0.034. That figure is a *diagnostic, not an alternative primary p-value*, and with only 29 possible shifts 1/29 is the finest resolution it can produce. No permutation *p*-value is attached to the detrended ρ at all, by design, because permuting individual residual years would reintroduce the exchangeability assumption the check exists to interrogate.

5. Discussion

Among the three salmon-temperature hypotheses, the Chinook association is the largest estimate and the most thoroughly checked, though it did not meet the prespecified support criterion. A1 is stable across all seven prespecified robustness checks, including all 29 leave-one-year-out reruns, and its estimated magnitude — roughly a fifth fewer fitted adults per standard deviation of warming — would be biologically meaningful if it were causal. Its Holm-adjusted *p* of 0.095 nevertheless fails the rule fixed in advance. The accurate description is a stable association that these data cannot confirm. Both halves of that sentence carry weight: omitting the first understates a consistent pattern, and omitting the second overstates the evidence.

How much the null results constrain is limited by design. A simulation using only the sample size and the prespecified alpha — reading no outcome data — indicates that detecting an association at the *strictest* Holm step within a three-test family would have required $|\rho| \approx 0.57$ for an 80% chance of detection, with roughly 39% power against a moderate $|\rho| = 0.40$. These figures describe that specific step under the simulation's bivariate-normal assumptions rather than an exact minimum detectable effect for the whole procedure. They do not reveal the true effect size or explain why A1 missed the threshold; what they do show is that this design could only reliably detect strong associations, so the three null results place weak bounds on the effects they failed to detect and are not evidence that those relationships are absent.

The two Coho estimates were weak and imprecise, and no comparison between species was formally tested. The Chinook estimate is numerically larger than either Coho estimate, but a difference between correlation estimates was not itself assessed, so this report makes no claim that the species respond differently. One untested possibility is that the fall Coho migration window is the coolest of the three, averaging about 11.7 °C, which could imply less thermal exposure than Chinook encounter a month

earlier; thermal stress was never measured, so that remains speculation rather than a result. A second consideration is biological rather than statistical: Issaquah Coho are hatchery yearlings reared in ponds rather than fish rearing freely in the creek for a year, so the exposure pathway assumed by A5 is likely weaker than the hypothesis presumed, though pond and in-creek thermal exposure were not compared and the hatchery does draw creek water.

Within the mechanism family the outcome was mixed, and describing it as "the physical chain held" would overstate it. Higher summer flow was associated with cooler modeled temperature, but April snowpack at a Cascade-crest station was not informative for late-summer flow in this lowland basin. Only one of the two links was supported, so the full pathway was not observed. The temperature-and-flow model explains a limited fraction of year-to-year variation in returns — adjusted R^2 of 0.126 for Chinook and below zero for Coho — which indicates that these particular measured predictors account for little of the variance in this fitted model. It does not identify which factor dominates. Ocean survival, harvest, hatchery release numbers, predation, and cohort structure are all unmeasured here, and at least one of them could plausibly move returns more than any variable in this analysis.

The most transferable lesson concerns the structure of the original question. The chain from snowpack to flow to temperature to salmon looks like a single hypothesis but joins two different kinds of claim. Its first half concerns physical watershed processes: directly measurable and governed by hydrology. Its second half concerns adult return abundance, which is mediated by ocean conditions, harvest, hatchery operations, and cohort structure — almost none of which appear in a freshwater dataset. The results split along that seam. The physical question was answerable with data a local watershed already collects; the demographic question was not resolved at this sample size, and the design sensitivity indicates the study had a low probability of reliably detecting a moderate effect even if one exists. That points toward longer records or measurements of the intervening life stages, rather than further re-analysis of these twenty-nine years.

6. Limitations

Design. This is an observational study of associations. No experiment was conducted, and no confounder is controlled beyond the linear calendar-year trend examined in §4.4 and the matched summer flow entered in the secondary descriptive model of §4.3. Nothing reported here supports a causal interpretation.

Measurement error in the exposure. Temperature is modeled rather than measured, and the error is not small relative to the signal. Held-out daily error inside the biological windows is 0.70–0.97 °C, whereas the between-year standard deviation of the window means the hypotheses actually use is only 0.53–0.54 °C. Averaging across a 47-day window (122 days for A5) reduces that error only to the extent daily errors

are independent, and they are not: the model shares inputs and can be biased across an entire season. Critically, **the uncertainty of the annual window means was never directly estimated**, so the effective exposure error is unknown and could be a substantial share of the variation being tested. Classical nondifferential measurement error often attenuates associations toward zero, but that behaviour is not guaranteed for rank correlations with model-derived error, so the direction of any bias here is not established. A separate caveat is that the ridge penalty was tuned using the same leave-one-year-out structure used to report accuracy, making the quoted errors mildly optimistic.

Sample size and temporal dependence. Twenty-nine return years is the binding constraint on the entire study, and it cannot be enlarged within this record. As §5 describes, the design had limited power against moderate associations, so null results bound the underlying effects only weakly. Years are also not fully exchangeable: both series trend with time, and the detrending and circular-shift checks are sensitivity diagnostics that cannot rule out nonlinear or cohort-scale temporal confounding.

Unmeasured confounding and administrative data. Several plausible drivers of adult returns are entirely absent: hatchery release numbers, harvest rates, life-stage-matched ocean conditions, predation at the Ballard Locks, and basin imperviousness. Hatchery release records are plausibly the most consequential omission, though no omitted variable was measured and their relative importance is untested — if fewer smolts were released in some years, adult returns two to four years later would move for reasons unrelated to freshwater conditions, and this analysis cannot separate the two. An access request to the standard regional source (PSMFC Regional Mark Information System) was declined for lack of an authorization key. Additionally, the salmon counts are administrative records maintained across three decades; no check for changes in counting procedure over time was performed.

Proxy scale and biological pathway. Stampede Pass April SWE is a regional indicator rather than a measure of Issaquah basin snowpack. Section 4.2 reports that it was not informative for Issaquah late-summer flow, but no basin-scale alternative was tested, so the A6 null cannot be separated from the limitations of this particular proxy. The A5 exposure pathway is weakened by the hatchery-yearling rearing practice described in §5. Hatchery- and wild-origin fish are pooled in the outcome because the origin-comparability audit was deferred.

Prespecification and multiplicity. The Phase 7 freeze is not a fully prospective preregistration, because earlier project phases had already examined generic annual relationships. The two-family multiplicity structure is itself a design limitation: it is less stringent than a single five-test family, and A7's classification as supported depends on that choice (§4.2).

7. Future work and reproducibility

Three next steps would address the largest uncertainties in this study, in order of value. **First, deploy a continuous in-stream temperature logger.** This is the most direct way to remove the study's central measurement limitation for future years, replacing a modeled exposure with a measured one and making true daily maxima and 7DADMax metrics available; obtaining the existing multi-decade Issaquah temperature records referenced in §1.1 is the complementary route for the historical period. It is worth noting that a logger improves future records but cannot retrospectively validate the historical window means used here; its value is prospective. **Second, obtain hatchery release records** through authorized RMIS access or an export via FISH, which would allow what is plausibly the most consequential omitted variable to be controlled — though, as §6 notes, the relative importance of the omitted variables was never tested. **Third, test the frozen A1 specification out of sample** on a comparable Lake Washington tributary. Independent replication is substantially stronger evidence than any further specification search within these same twenty-nine years. A basin-appropriate hydrologic variable — baseflow recession or groundwater level in place of Cascade-crest SWE — would also test the water-availability idea properly.

Reproducibility. The analysis is built on public data with a machine-readable audit trail. Raw downloads are stored with checksums, cleaned into standardized annual tables, and joined into a single analysis table, with no later stage editing an earlier one. The Phase 7 analysis was executed under a protocol committed and hashed before any of its own life-stage tests was run; its outputs are recorded in a freeze record listing a SHA-256 for every result file, and all randomness uses fixed seeds, so the permutation and bootstrap values are exactly reproducible. If a software error is ever found, the protocol requires preserving the original package and writing a separately versioned replacement rather than overwriting it.

A reviewer found in September 2026 that a freshly downloaded copy of the repository failed its own integrity check, and the repository was repaired in response. Git's end-of-line conversion on Windows rewrote file bytes at checkout, which changed the SHA-256 of frozen text artifacts even though their contents were identical — the substantive result tables always reproduced correctly, and the failure was one of packaging rather than of the analysis. This was fixed, and a clean clone of the repaired repository now passes the independent Phase 7 validator on all 28 of its checks, with all 24 frozen artifact hashes verifying. Those repairs are committed locally and published to the project's public repository.

ASSISTANCE AND AI USE — FULL DISCLOSURE

Across the project, AI (Claude) assistance was used for initial code implementation, for reviewing code, analysis pipeline, simulation results and research report, for locating candidate literature sources, and for audit report prose formatting and style. The research question, the originating observations, the hypotheses, the life-

stage window definitions, the frozen protocol design, and the decisions about what may and may not be claimed are the student's own. Every numerical result reported here comes from the frozen, independently validated analysis package rather than from any language model summary of it. Every statistic in Tables 1–3 and Figures 1–4 comes from the frozen analysis package rather than from any language model summary of it. The one exception is the design-sensitivity calculation quoted in §5, which was computed after the freeze from the sample size and prespecified alpha alone; it reads no outcome data and leaves the frozen package untouched. The package structure, hashes, and multiplicity arithmetic were checked by an independent output validator, which tests the outputs for structural and internal consistency but does not recompute the association statistics from source data.

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